DEPARTMENT OF TELECOMMUNICATION ENGINEERING
BACHELOR OF SCIENCE IN CYBER SECURITY
MEHRAN UNIVERSITY OF ENGINEERING & TECHNOLOGY, JAMSHORO
NETWORK SECURITY (CYS360)
(6^{TH} SEMESTER, 3^{RD} Year) LAB EXPERIMENT # 09

Name:	**___Muhammad_Usman______**			**Roll number:**	**__23BSCYS006______**
Score:	**__________**	**Date:**	**____________**	**Instructor's Signature:**	**______________**

Lab # 09: To configure Maximum and Guaranteed Bandwidth for Bandwidth Management on Enterprise Networks

Objective: The objective of this lab is to set overall maximum bandwidth to restrict non-key service traffic on an enterprise network and setting overall guaranteed bandwidth to ensure proper forwarding of key service traffic during peak hours.

Duration: 3 Hours

Expected Outcomes: By the end of this lab session, students will be able to: 1.
Restrict network traffic
2. Set maximum traffic bandwidth for different applications such as p2p, email and enterprise resource planning (ERP).

Rubrics

LAB PERFORMANCE INDICATOR	SUBJECT KNOWLEDGE	DATA ANALYSIS AND INTERPRETATION	ABILITY TO CONDUCT EXPERIMENT
SCORE			

Note: Student performance will be assessed based on this rubric during lab sessions.

Networking Requirements

As shown in Figure 1, an enterprise purchases 100 Mbit/s bandwidth from an ISP. On office networks, email and ERP traffic is key service traffic, and P2P and online video traffic is nonkey service traffic. However, P2P and online video traffic exhausts the limited bandwidth resources on the enterprise network, and key service traffic, such as email and ERP traffic, is not properly forwarded. As a result, emails fail to be sent, and web pages fail to be displayed, which greatly affects the daily operation of the enterprise.

To prevent the preceding symptoms, the enterprise requires to enable the bandwidth management function on the FW to meet the following requirements:

- Restrict P2P and online video traffic within 30 Mbit/s at any time. Restrict P2P and online video connections within 10,000.
- Assign a minimum of 60 Mbit/s bandwidth for applications, such as email and ERP.

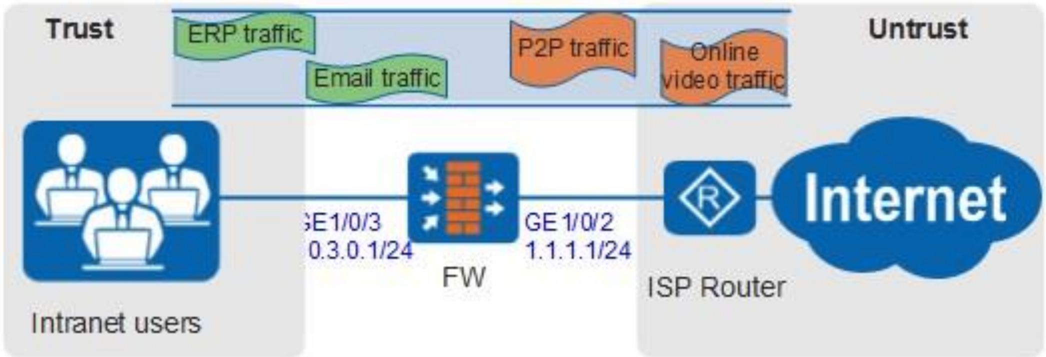


Figure 1 Networking diagram for configuring the maximum and guaranteed bandwidth for bandwidth management on enterprise networks

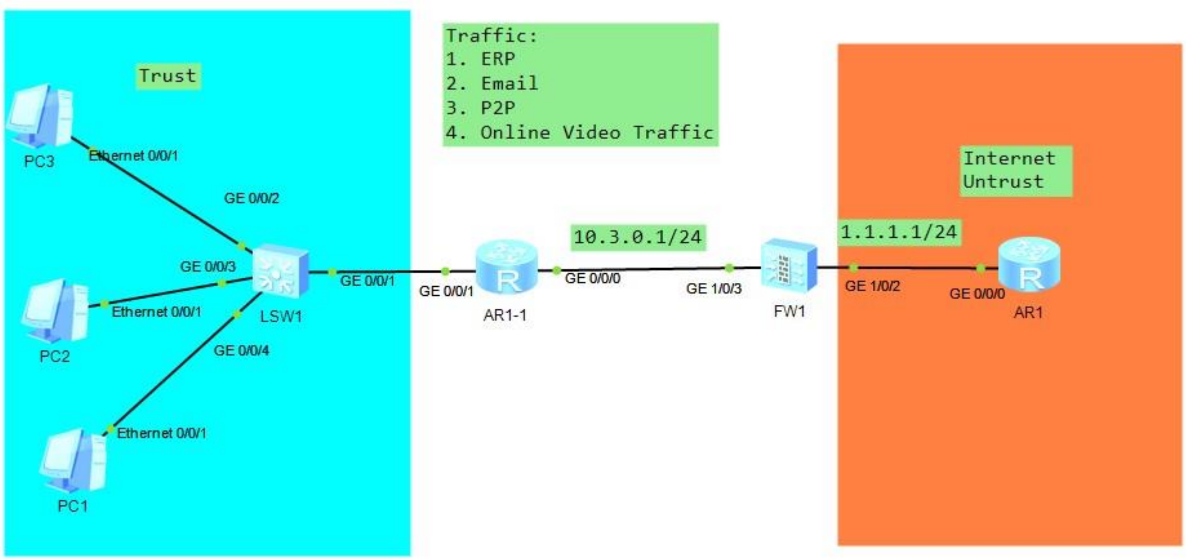


Figure 2 Networking diagram for Traffic Management

Configuration Roadmap

1. Set interface IP addresses and assign the interfaces to security zones.
2. Configure a traffic policy for P2P and online video applications and reference the traffic profile in which the overall maximum bandwidth is 30 Mbit/s and overall maximum number of connections is 10,000.

3. Configure a traffic policy for email and ERP applications and reference the traffic profile in which the overall guaranteed bandwidth is 60 Mbit/s.

NOTE:

- Upstream and downstream depend on the direction of FW bandwidth policy. For simplicity, upstream refers to the direction from Trust to Untrust, and downstream refers to Untrust to DMZ in this section.
- Assuming that the security zones, routers, and security policies have been configured, this section introduces only how to configure bandwidth management.

Procedure

1. Set interface IP addresses and assign the interfaces to security zones.

   a. Set an IP address for interface GigabitEthernet 1/0/2 and assign the interface to the untrust zone.

```
<FW> system-view
[FW] interface GigabitEthernet 1/0/2
[FW-GigabitEthernet1/0/2] ip address 1.1.1.1 24
[FW-GigabitEthernet1/0/2] quit
[FW] firewall zone untrust
[FW-zone-untrust] add interface GigabitEthernet 1/0/2
[FW-zone-untrust] quit
```

   b. Set an IP address for interface GigabitEthernet 1/0/3 and add the interface to the trust zone.

```
[FW] interface GigabitEthernet 1/0/3
[FW-GigabitEthernet1/0/3] ip address 10.3.0.1 24
[FW-GigabitEthernet1/0/3] quit
[FW] firewall zone trust
[FW-zone-trust] add interface GigabitEthernet 1/0/3
[FW-zone-trust] quit
```

2. Configure a schedule.

```
[FW] time-range work_time
[FW-time-range-work_time] period-range 09:00:00 to 18:00:00 working-day
[FW-time-range-work_time] quit
```

3. Configure a traffic profile for P2P and online video services.

```
[FW] traffic-policy
[FW-policy-traffic] profile profile_p2p
[FW-policy-traffic-profile-profile_p2p] bandwidth maximum-bandwidth whole both 30000
[FW-policy-traffic-profile-profile_p2p] bandwidth connection-limit whole both 1000
[FW-policy-traffic-profile-profile_p2p] quit
```

4. Configure a traffic policy for P2P and online video services.

NOTE:

The following example describes the bandwidth management configuration for BitTorrent (BT) and eDonkey/eMule P2P services. You can specify other P2P services as required.

```
[FW-policy-traffic] rule name policy_p2p
[FW-policy-traffic-rule-policy_p2p] source-zone trust
[FW-policy-traffic-rule-policy_p2p] destination-zone untrust
[FW-policy-traffic-rule-policy_p2p] application app BT YouKu
[FW-policy-traffic-rule-policy_p2p] action qos profile profile_p2p
[FW-policy-traffic-rule-policy_p2p] quit
```

5. Configure a traffic profile for email and ERP services.

```
[FW-policy-traffic] profile profile_email
[FW-policy-traffic-profile-profile_email] bandwidth guaranteed-bandwidth whole both
60000
[FW-policy-traffic-profile-profile_email] quit
```

6. Configure a traffic policy for email and ERP services.

NOTE:

The following example describes the bandwidth management configuration for Outlook Web Access and LotusNotes. You can specify other P2P services as required.

```
[FW-policy-traffic] rule name policy_email
[FW-policy-traffic-rule-policy_email] source-zone trust
[FW-policy-traffic-rule-policy_email] destination-zone untrust
[FW-policy-traffic-rule-policy_email] application app LotusNotes OWA
[FW-policy-traffic-rule-policy_email] time-range work_time
```

```
[FW-policy-traffic-rule-policy_email] action qos profile profile_email [FW-
policy-traffic-rule-policy_email] quit
```

Task:

1. Attach Screenshots of all the topology and confinguration.

Note: Attach all the screenshots of the configuration. The Firewall name should be your roll number like FW1-22BSCYS001. Upload the .topo and .cfg files on MS Teams Assignment.

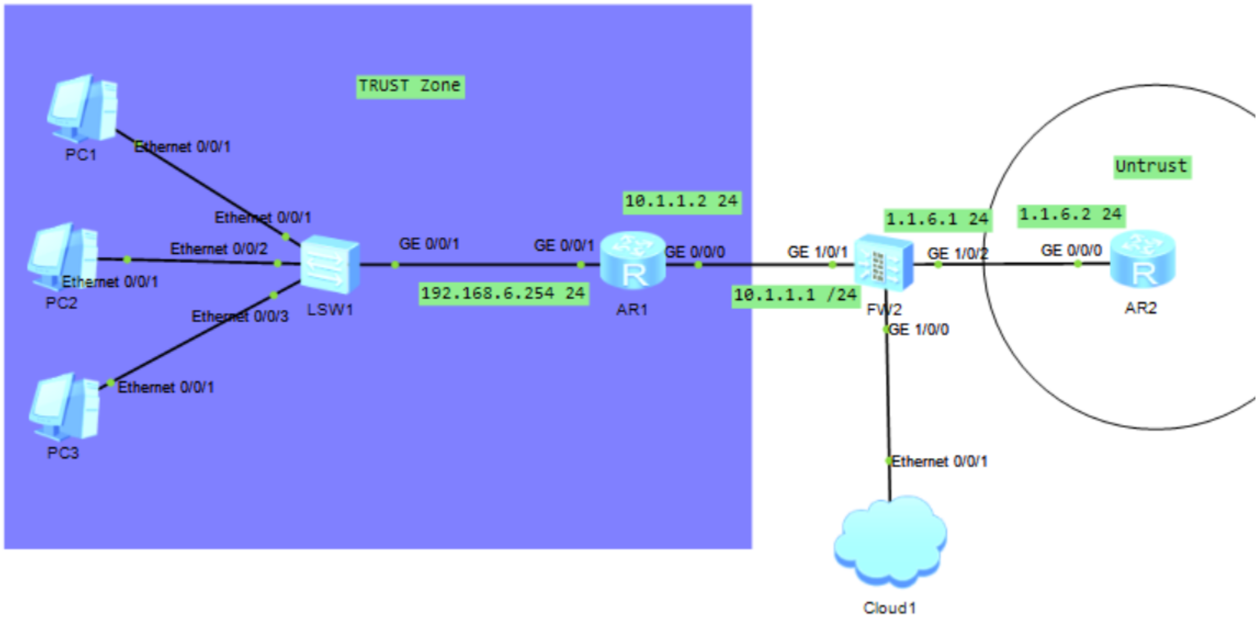
Questions:

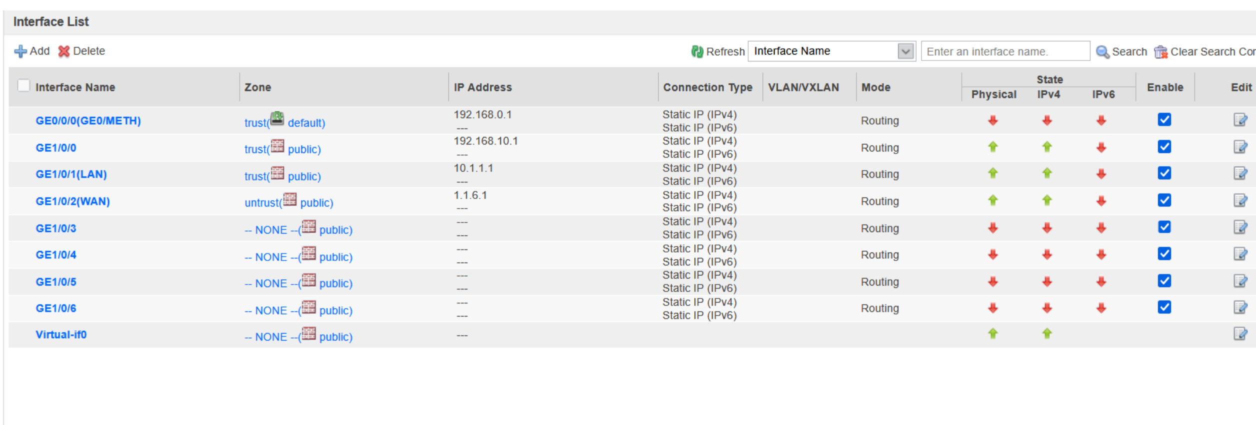
1. **Bandwidth management is necessary** to ensure critical business applications get enough network resources and to prevent congestion caused by non-essential traffic.
2. **Maximum bandwidth** is the highest amount of bandwidth an application can use. **Guaranteed bandwidth** is the minimum amount reserved for an application to ensure stable performance.
3. **Email and ERP** are considered key traffic because they are essential for daily business operations and productivity.

4. If total demand exceeds **100 Mbit/s**, P2P and online video traffic will be limited, throttled, or dropped first to protect critical applications.

5. If Email and ERP need **70 Mbit/s**, the firewall will prioritize and guarantee that 70 Mbit/s for them, and the remaining 30 Mbit/s will be shared among lower-priority traffic.

6. A **firewall (FW)** controls and prioritizes network traffic, applies bandwidth limits, guarantees minimum bandwidth for critical applications, and prevents non-essential traffic from overwhelming the network.

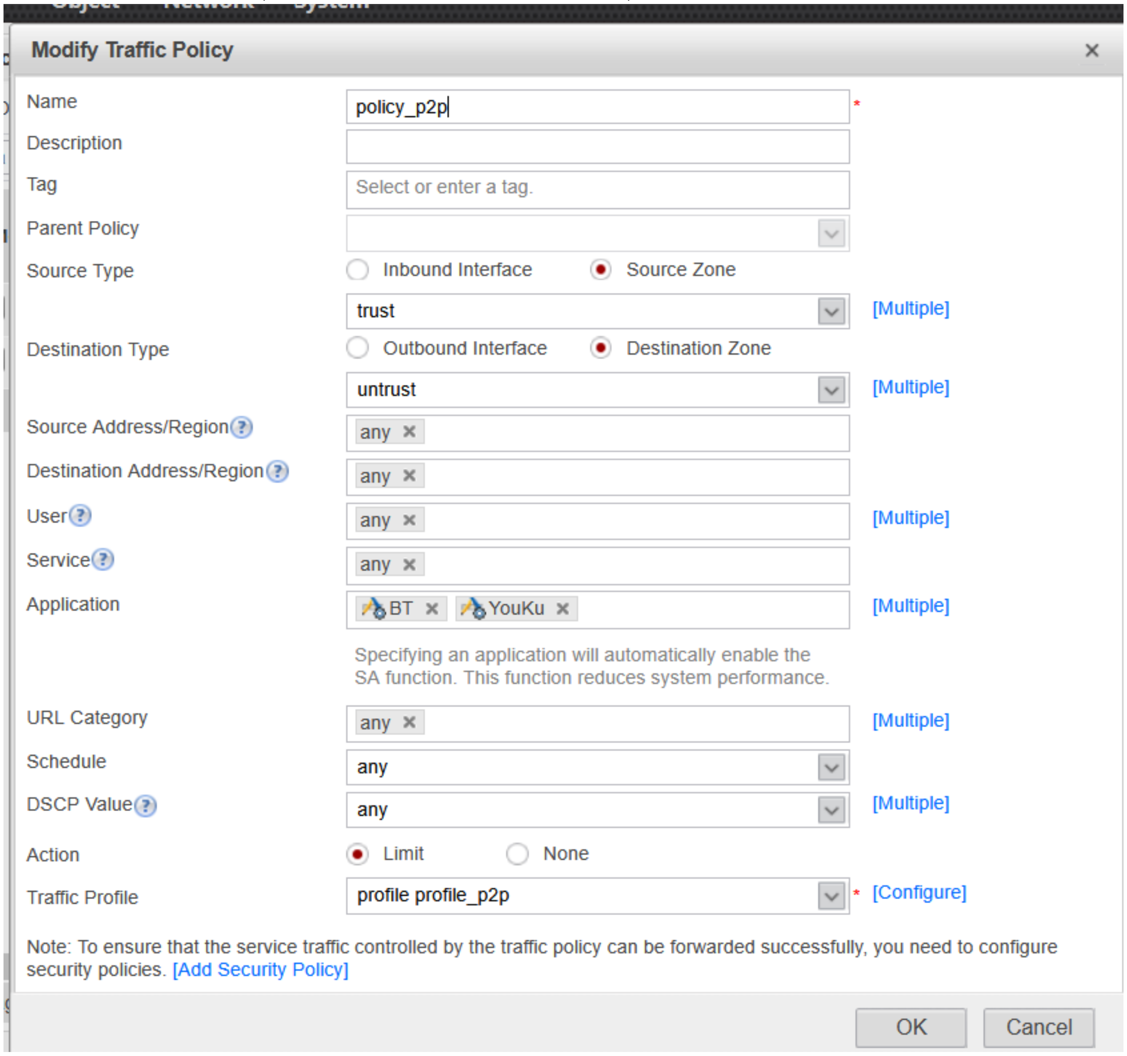


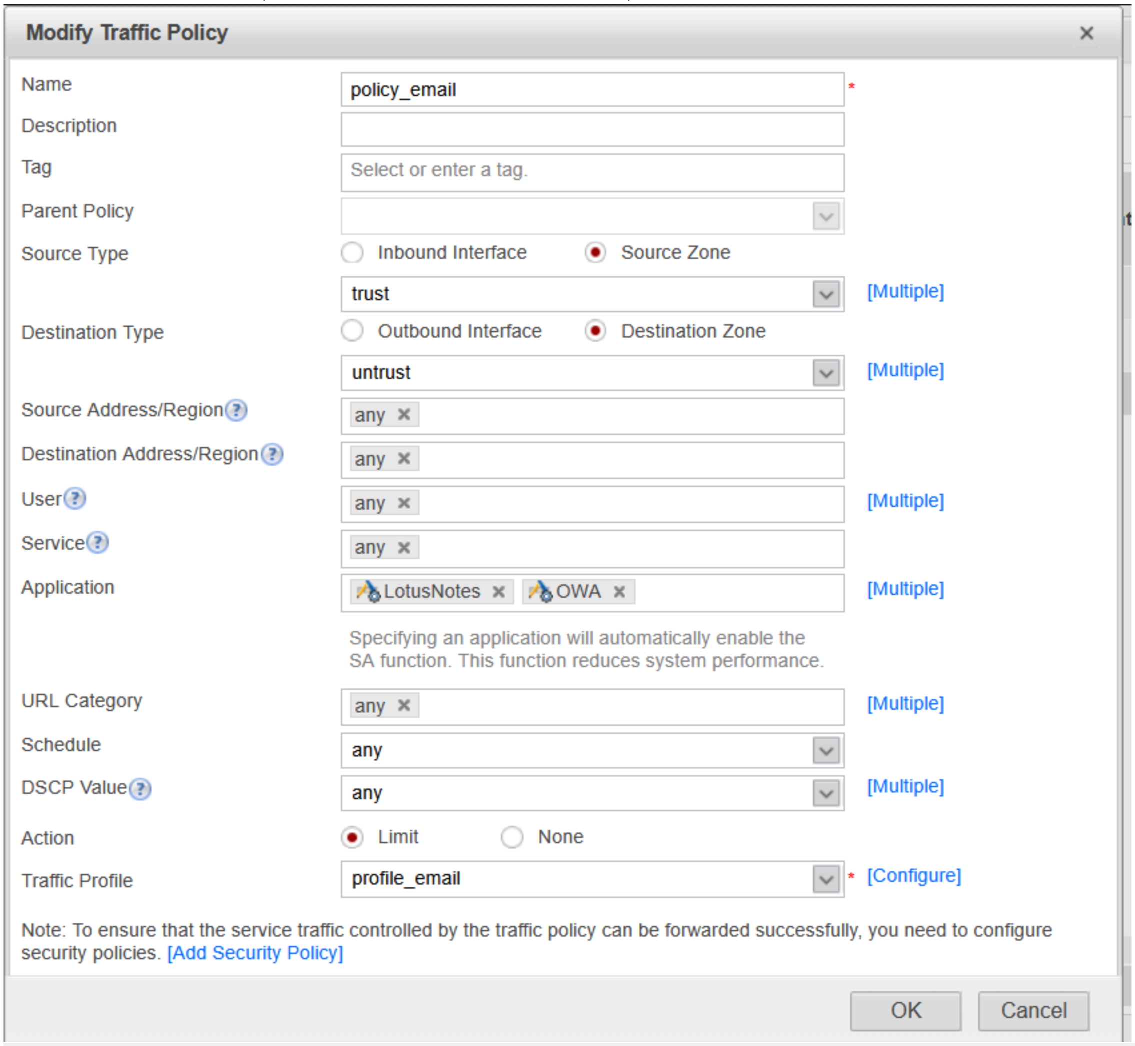


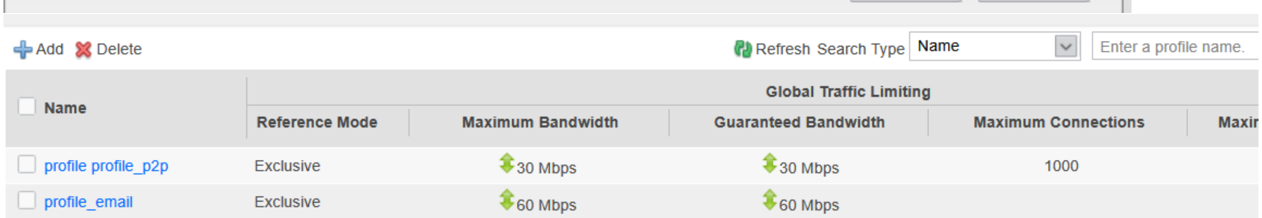
Interface List

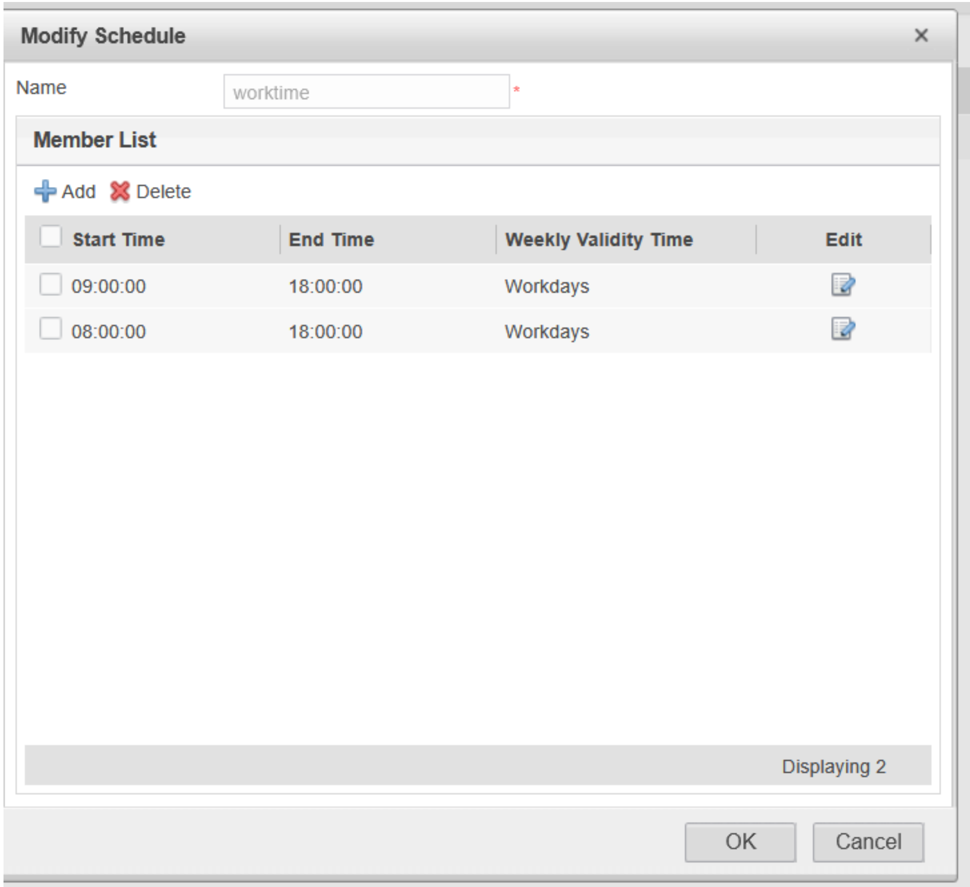
Add Delete Refresh Interface Name Enter an interface name. Search Clear Search Con

Interface Name	Zone	IP Address	Connection Type	VLAN/VXLAN	Mode	State: Physical	State: IPv4	State: IPv6	Enable	Edit
GE0/0/0(GE0/METH)	trust(default)	192.168.0.1 ---	Static IP (IPv4) Static IP (IPv6)		Routing	↓	↓	↓	☑	
GE1/0/0	trust(public)	192.168.10.1 ---	Static IP (IPv4) Static IP (IPv6)		Routing	↑	↑	↓	☑	
GE1/0/1(LAN)	trust(public)	10.1.1.1 ---	Static IP (IPv4) Static IP (IPv6)		Routing	↑	↑	↓	☑	
GE1/0/2(WAN)	untrust(public)	1.1.6.1 ---	Static IP (IPv4) Static IP (IPv6)		Routing	↑	↑	↓	☑	
GE1/0/3	-- NONE --(public)	--- ---	Static IP (IPv4) Static IP (IPv6)		Routing	↓	↓	↓	☑	
GE1/0/4	-- NONE --(public)	--- ---	Static IP (IPv4) Static IP (IPv6)		Routing	↓	↓	↓	☑	
GE1/0/5	-- NONE --(public)	--- ---	Static IP (IPv4) Static IP (IPv6)		Routing	↓	↓	↓	☑	
GE1/0/6	-- NONE --(public)	--- ---	Static IP (IPv4) Static IP (IPv6)		Routing	↓	↓	↓	☑	
Virtual-if0	-- NONE --(public)	---				↑	↑			


Modify Traffic Policy
Name
policy_p2p
Description
Tag
Select or enter a tag.
Parent Policy
Source Type
Inbound Interface
Source Zone
trust
[Multiple]
Destination Type
Outbound Interface
Destination Zone
untrust
[Multiple]
Source Address/Region
any
Destination Address/Region
any
User
any
[Multiple]
Service
any
Application
BT
YouKu
[Multiple]
Specifying an application will automatically enable the SA function. This function reduces system performance.
URL Category
any
[Multiple]
Schedule
any
DSCP Value
any
[Multiple]
Action
Limit
None
Traffic Profile
profile profile_p2p
[Configure]
Note: To ensure that the service traffic controlled by the traffic policy can be forwarded successfully, you need to configure security policies. [Add Security Policy]
OK
Cancel


Modify Traffic Policy
Name
policy_email
Description
Tag
Select or enter a tag.
Parent Policy
Source Type
Inbound Interface
Source Zone
trust
[Multiple]
Destination Type
Outbound Interface
Destination Zone
untrust
[Multiple]
Source Address/Region
any
Destination Address/Region
any
User
any
[Multiple]
Service
any
Application
LotusNotes
OWA
[Multiple]
Specifying an application will automatically enable the SA function. This function reduces system performance.
URL Category
any
[Multiple]
Schedule
any
DSCP Value
any
[Multiple]
Action
Limit
None
Traffic Profile
profile_email
[Configure]
Note: To ensure that the service traffic controlled by the traffic policy can be forwarded successfully, you need to configure security policies. [Add Security Policy]
OK
Cancel


Add
Delete
Refresh
Search Type
Name
Enter a profile name.
Name
Global Traffic Limiting
Reference Mode
Maximum Bandwidth
Guaranteed Bandwidth
Maximum Connections
profile profile_p2p
Exclusive
30 Mbps
30 Mbps
1000
profile_email
Exclusive
60 Mbps
60 Mbps


Modify Schedule
Name
worktime
Member List
Add
Delete
Start Time
End Time
Weekly Validity Time
Edit
09:00:00
18:00:00
Workdays
08:00:00
18:00:00
Workdays
Displaying 2
OK
Cancel